IT23302 | DATA STRUCTURES

MINI PROJECT PROPOSAL

Smart Student Record & Search System

A C-based console program using a Binary Search Tree, Queue, searching and sorting techniques.

PROJECT TEAM

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Submission Date: 28.10.2026

Abstract

Keeping track of student information becomes difficult when records are handled manually, especially when staff need to locate a particular student or compare marks across a large group. The Smart Student Record & Search System is proposed as a simple console application written in C to make common record-management tasks more organised and demonstrate the practical use of data structures.

The program will maintain each student's register number, name, department and marks. A Binary Search Tree (BST) will organise records by register number and provide operations for adding, finding and removing records. Its inorder traversal will present records in ascending register-number order. A Queue will be used for students waiting to be verified or processed, ensuring that requests are handled in the order they arrive through the First In, First Out (FIFO) rule.

Searching and sorting methods will help users retrieve information and arrange student records according to marks, names or register numbers. The proposed system can also include checks for duplicate register numbers and invalid marks. If permitted within the project scope, file operations can be added to preserve records between program sessions. The project is intended to show how choosing suitable data structures can make a small record-management application clearer, more structured and easier to maintain.

Project Objectives

• Create a menu-based C application for maintaining basic student details.

• Use a BST to organise student records by register number and support insertion, lookup, deletion and traversal.

• Apply a queue to handle student verification or processing requests in arrival order.

• Provide searching options to find a student by register number and optionally by name.

• Arrange records using a sorting method, such as bubble sort, according to marks or another chosen field.

• Check user input and reject duplicate register numbers or marks outside the allowed range.

• Explore file storage as an extension so saved records can be loaded during a later run.

• Observe the efficiency and limitations of the selected data structures and algorithms.

Data Structures and Algorithms

Technique	How it is applied
C structure	Groups each student's register number, name, department and marks into a single record.
Binary Search Tree	Uses register number as the key for inserting, locating and deleting records; inorder traversal lists keys in ascending order.
Queue	Holds students awaiting processing. Enqueue adds a request and dequeue removes the request at the front.
Searching	BST search is used for register numbers; a linear search can be added for names.

Technique	How it is applied
Sorting	A method such as bubble sort orders records by marks, name or register number.
File operations (extension)	Can write records to a file and read them back when the program is started again.

System Design

The BST will be the main structure used to organise and locate student records by register number. The queue will serve a different purpose: tracking the order in which students are waiting for an action. Search and sort functions will operate on the stored records, while validation checks will help prevent inconsistent entries. File storage is an optional extension and should be included only if it is implemented in the final program.

Planned Modules

Add a record • View all records • Find a student • Edit student details • Remove a record • Sort the list • Add a student to the processing queue • Process the next student • View pending queue entries • Validate input • Optional save/load feature

Expected Result

The completed program is expected to provide a straightforward way to maintain and retrieve student details through a text-based menu. It will demonstrate the BST's ordered record operations, the queue's FIFO behaviour, and the use of searching and sorting methods. The project will also give practical experience in writing modular C functions and considering the efficiency of different operations.

Tools and Technologies

Programming language: C **Compiler:** GCC or another C compiler
Interface: Text-based console
Data storage: Memory; file storage as an optional extension